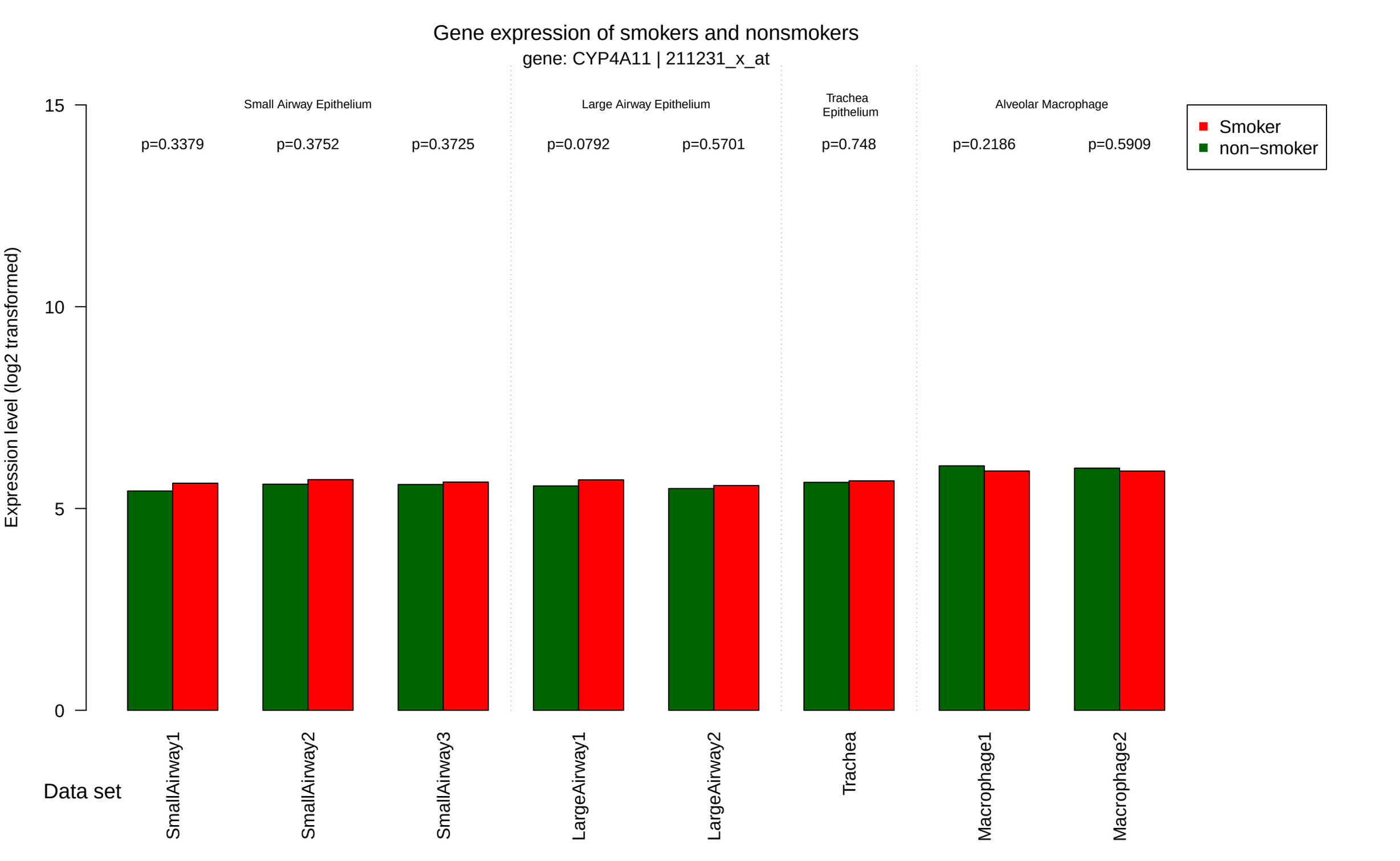
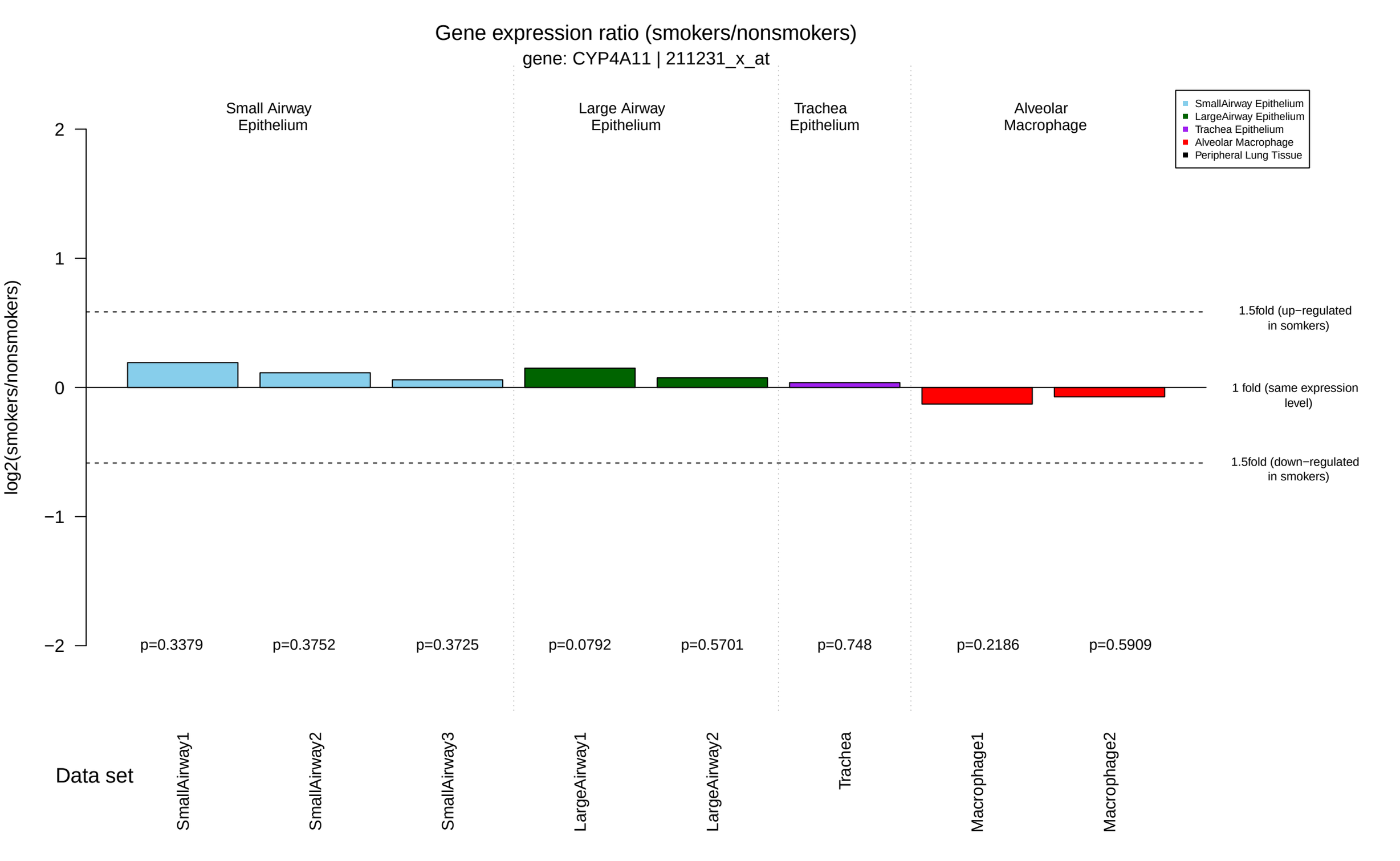
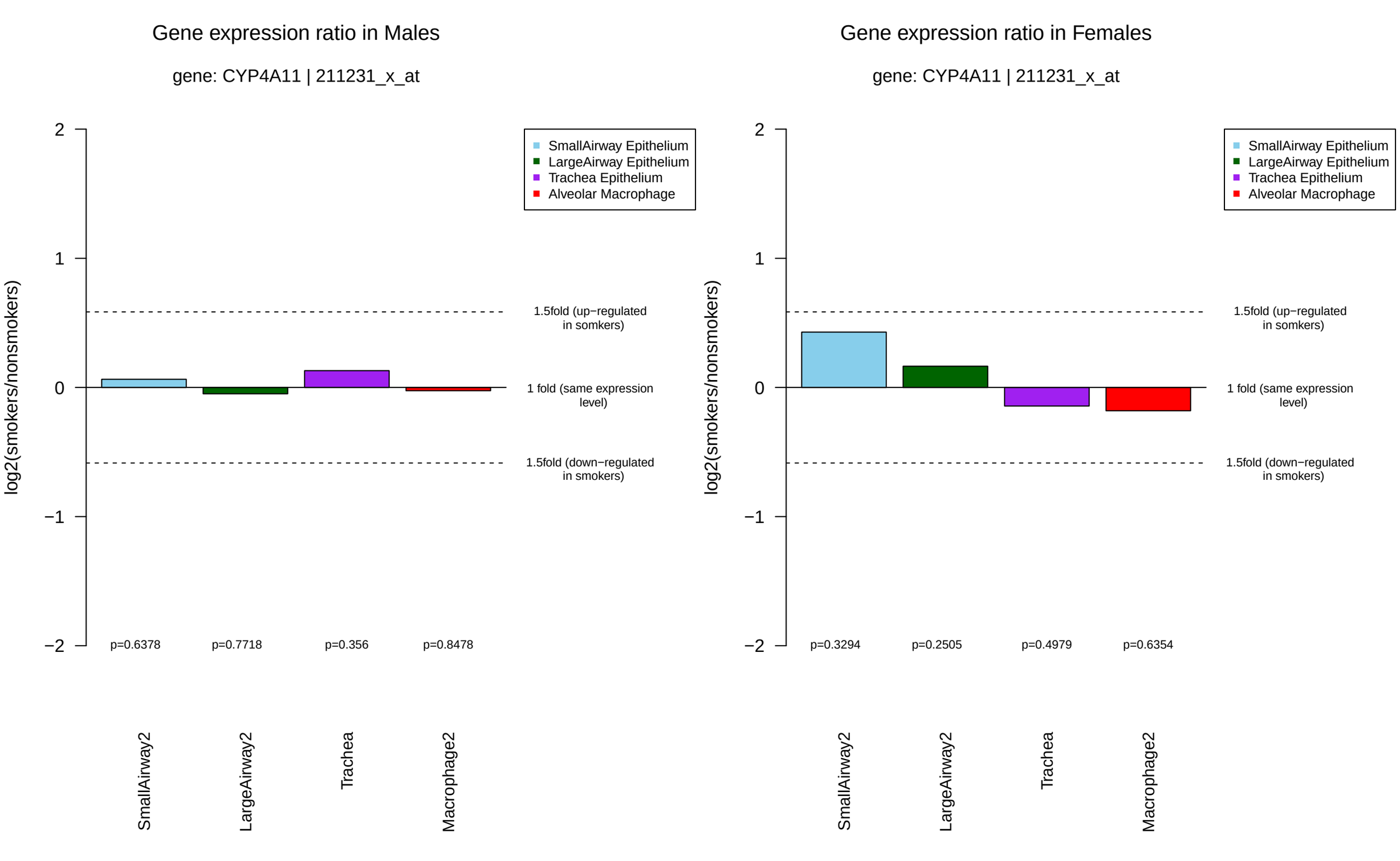
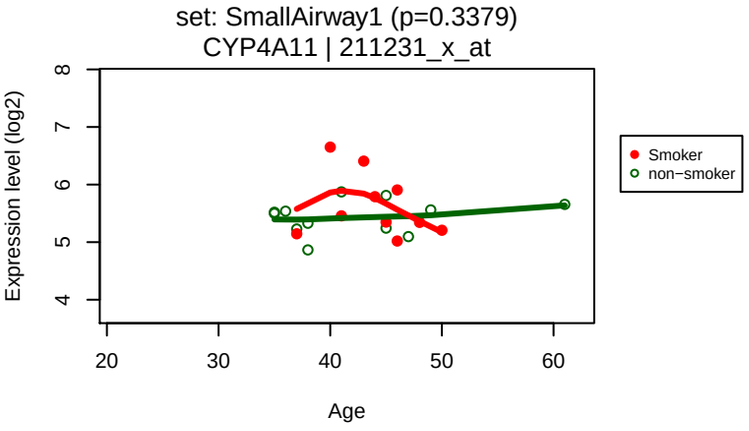
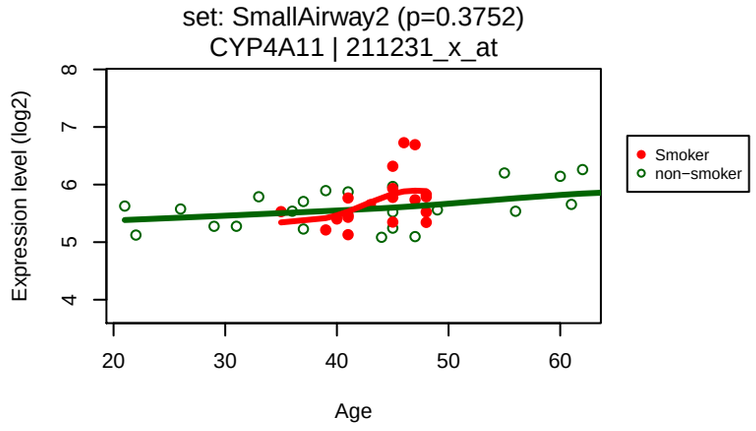
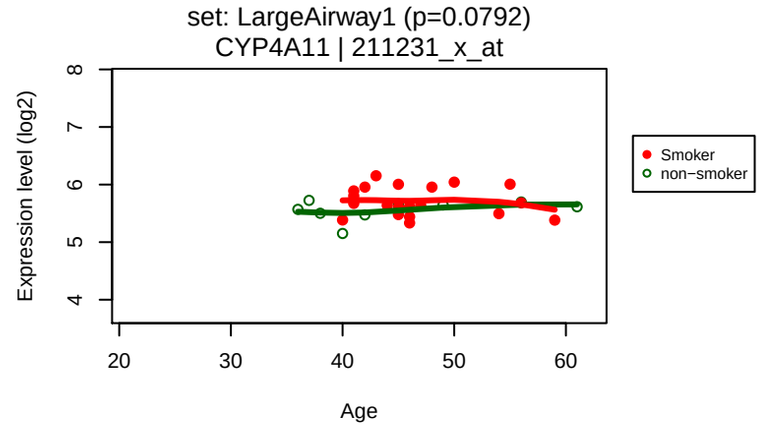

Gene expression of smokers and nonsmokers
gene: CYP4A11 | 211231_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3379
p=0.3752
p=0.3725
p=0.0792
p=0.5701
p=0.748
p=0.2186
p=0.5909
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

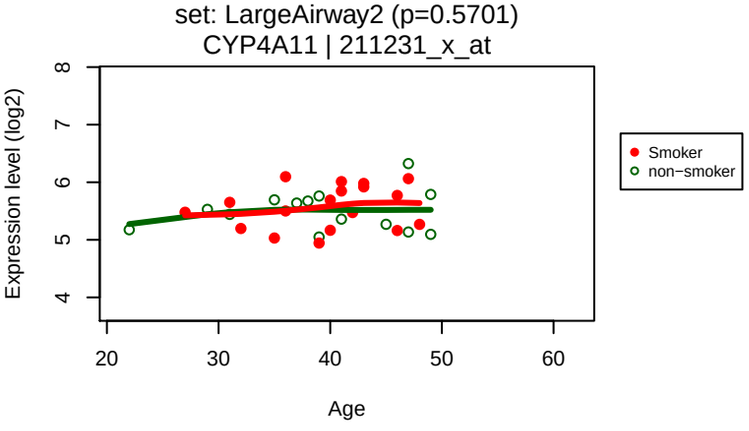

Gene expression ratio (smokers/nonsmokers)
gene: CYP4A11 | 211231_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3379
p=0.3752
p=0.3725
p=0.0792
p=0.5701
p=0.748
p=0.2186
p=0.5909
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

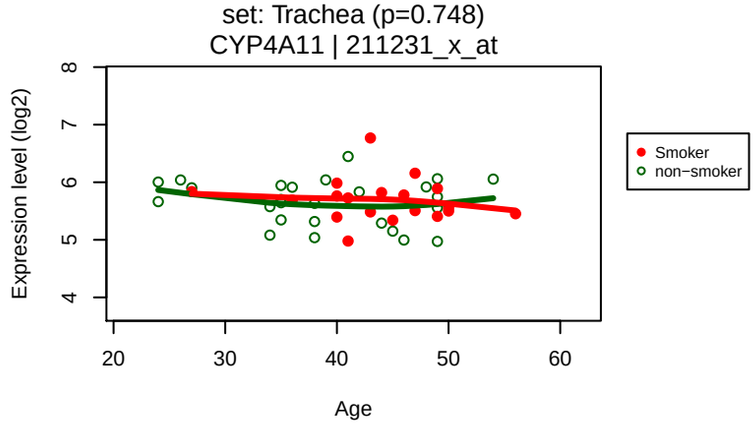

Gene expression ratio in Males
gene: CYP4A11 | 211231_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6378
p=0.7718
p=0.356
p=0.8478
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP4A11 | 211231_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3294
p=0.2505
p=0.4979
p=0.6354
SmallAirway2
LargeAirway2
Trachea
Macrophage2

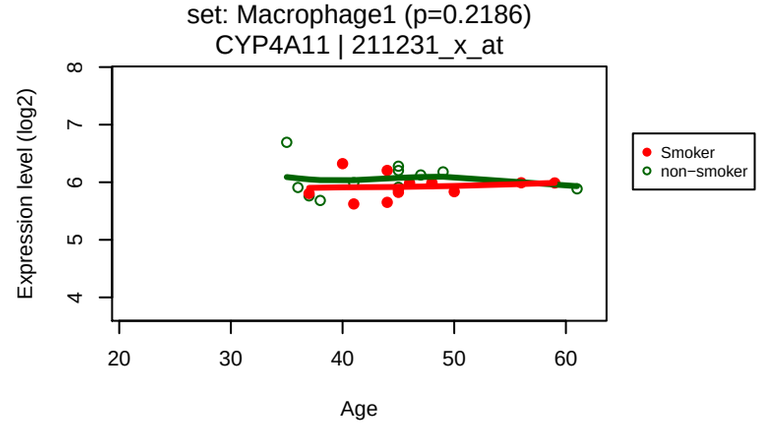

set: SmallAirway1 (p=0.3379)
CYP4A11 | 211231_x_at
Expression level (log2)
Age
Smoker
non−smoker

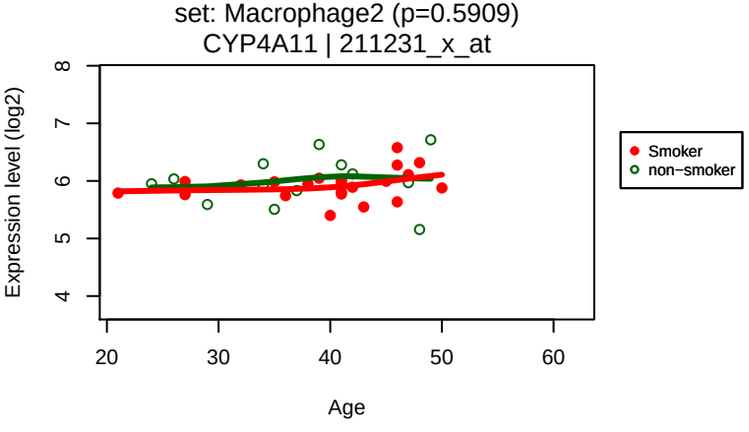

set: SmallAirway2 (p=0.3752)
CYP4A11 | 211231_x_at
Expression level (log2)
Age
Smoker
non−smoker

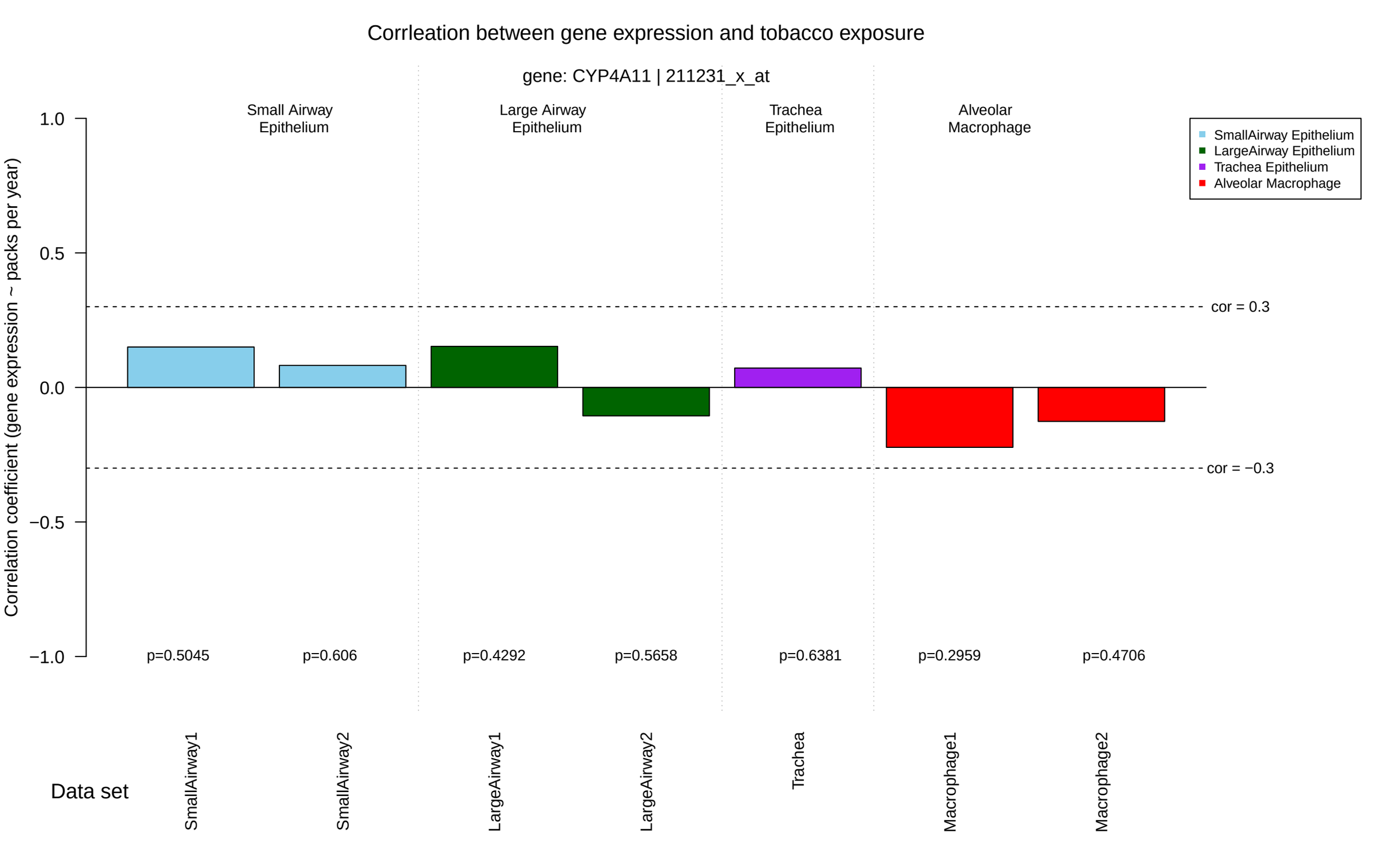

set: LargeAirway1 (p=0.0792)
CYP4A11 | 211231_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5701)
CYP4A11 | 211231_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.748)
CYP4A11 | 211231_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2186)
CYP4A11 | 211231_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5909)
CYP4A11 | 211231_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP4A11 | 211231_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5045
p=0.606
p=0.4292
p=0.5658
p=0.6381
p=0.2959
p=0.4706
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2